

Lab 7 - Thresholding

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1 Theory

Image Thresholding is a fundamental technique in digital image processing. It is used to create binary images from grayscale images by turning all pixels below a certain threshold to zero (black) and all pixels above that threshold to a maximum value (white). This process is especially useful for separating objects from the background.

Mathematically, for a given pixel intensity r and a threshold value T , the thresholding operation S can be defined as:

$$S = \begin{cases} 1, & \text{if } r \geq T \\ 0, & \text{if } r < T \end{cases}$$

where:

- r is the intensity of the pixel in the grayscale image,
- T is the threshold value,
- S is the output pixel value (binary: 0 or 1).

This operation converts the grayscale image into a binary image, highlighting regions of interest.

```
[1]: #import libraries
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
```

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[2]: def Threshold_Image(image):
    threshold = int(input("Enter Value of threshold"))
    image = image.resize((400,400), Image.Resampling.LANCZOS)
    # convert to numpy array
    numpy_image = np.array(image)

    row = numpy_image.shape[0]
    column = numpy_image.shape[1]

    new_array = np.zeros(shape=(row,column))

    for i in range(row):
        for j in range(column):
```

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    if(numpy_image[i][j]>=threshold):
        new_array[i][j] = 255
    else:
        new_array[i][j] = 0

    #converting array back to image
    threshold_image = Image.fromarray(new_array)
    threshold_image = threshold_image.convert("L")

    return threshold_image

```

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[3]: # reading image and converting to gray scale
image = Image.open('img/prakash.jpg').convert('L')
# calling negative function
threshold_image = Threshold_Image(image)

#displaying the images
fig = plt.figure()
fig.set_figheight(20)
fig.set_figwidth(20)

fig.add_subplot(1,2,1)
plt.imshow(image, cmap='gray')
plt.title('Original image')

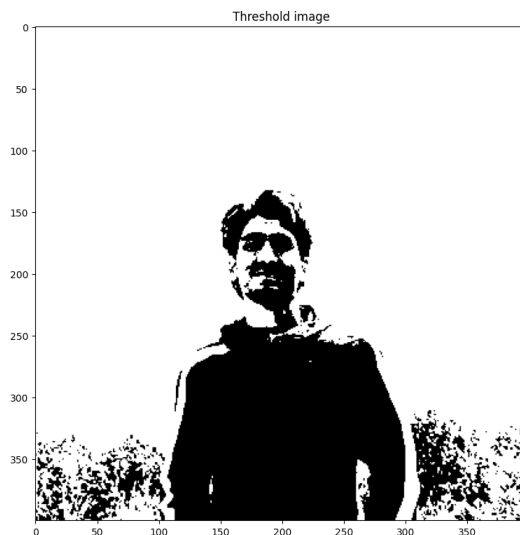
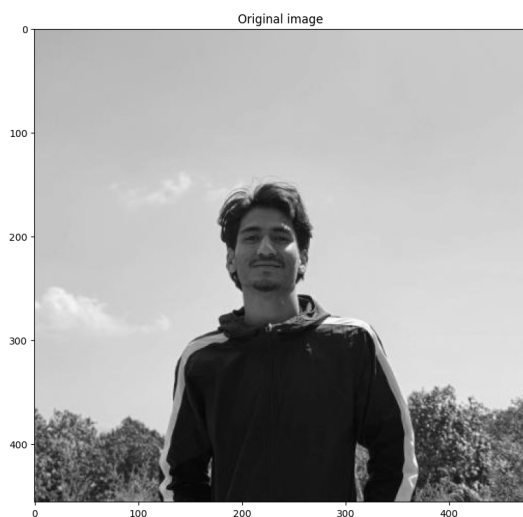
fig.add_subplot(1,2,2)
plt.imshow(threshold_image, cmap='gray')
plt.title('Threshold image')

```

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[3]: Text(0.5, 1.0, 'Threshold image')

```



2 Conclusion

From the above lab work, we implemented image thresholding using Python. The process involved:

- Reading and converting an image to grayscale,
- Applying a user-defined threshold to segment the image,
- Displaying both the original and thresholded images for comparison.

The thresholding technique effectively separates the foreground from the background, making it easier to analyze specific regions in an image. This method is widely used in applications such as document scanning, medical imaging, and object detection. The choice of threshold value T is crucial, as it directly affects the quality of segmentation. Proper selection of T can significantly enhance the visibility of important features in the image.